Advanced database management systems

Topic 4: Column store and partitioning

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Overview

- For high demand data warehouse applications, the relational databases should be reorganized
- We use column stores instead of row stores, and compress and partition large tables

After completing this topic:

- you will understand the pro's and con's of column stores and partitioning
- you will be able to design and implement a column store and partition it using MS SQL Server and Postgres technologies

The principle of a column store

- The same columns from all records are kept together
- The pages contain a singel column of many records
- Advantages
  - Most queries need only a small subset of all columns
  - Less data is to be moved into the data buffer -> fast queries
- Disadvantages
  - Records must be assembled from multiple column stores
  - INSERT/UPDATE/DELETE must modify all column stores -> slow modifications

Compression

- Data is being compressed/decompressed transparently
  - Advantage: less data to be moved, faster I/O
  - Disadvantage: computational overhead
- Row or page compression
- Page compression for column store
  - A column store of an attribute column has low cardinality
  - Low cardinality pages compress extremely well
  - Substantial speedup in SELECT queries (20x)

Partitioning

- The idea is divide and conquer
- For large tables that do not fit into memory
- A table can be partitioned according to a single **partitioning key**, e.g. date
- Improved performance at the expense of more resources and complexity
- Defined by a p. **scheme** and a p. **function**
- Loading: partitions can be swapped with minimally logged operations
- Can be combined with column store and compression